

Experiment – 4

Aim: To implement and execute advanced SQL queries on a relational database using joins, subqueries, set operations, aggregate functions, grouping, and conditional operators for effective data retrieval and analysis.

Q1. Find match Ids of all the matches in the database in which tendulkar batted.

USE aarya61_match

SELECT MatchID FROM Batting WHERE PID IN

(SELECT PlayerID FROM Player WHERE LName = 'Tendulkar');

```
MySQL localhost:33060+ ssl SQL > USE aarya61_match
Default schema set to 'aarya61_match'.
Fetching global names, object names from 'aarya61_match' for auto-completion...
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT MatchID
-> FROM Batting
-> WHERE PID IN
-> (SELECT PlayerID
-> FROM Player
-> WHERE LName = 'Tendulkar');
```

MatchID
2689

1 row in set (0.0019 sec)

Q2. Find match information of all matches in which Dhoni has batted.

SELECT Team1, Team2, Ground, Date FROM ODIMatch, Batting, Player

WHERE ODIMatch.MatchID = Batting.MatchID

AND Player.PlayerID = Batting.PID AND Player.LName = 'Dhoni';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT Team1, Team2, Ground, Date
-> FROM ODIMatch, Batting, Player
-> WHERE ODIMatch.MatchID = Batting.MatchID
-> AND PlayerID = PID
-> AND LName = 'Dhoni';
```

Team1	Team2	Ground	Date
Sri Lanka	India	Colombo	2008-08-24
Australia	India	Brisbane	2008-03-04

2 rows in set (0.0027 sec)

Q3. Find match Ids of all matches in which Tendulkar batted.

SELECT MatchID FROM Batting, Player WHERE PID=PlayerID AND LName='Tendulkar';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT MatchID
-> FROM Batting, Player
-> WHERE PID = PlayerID
-> AND LName = 'Tendulkar';
```

MatchID
2689

1 row in set (0.0016 sec)

Q4. Find Ids of all players that have both bowled and batted in the ODI match 2689

```
SELECT PID FROM Batting WHERE MatchID=2689
AND PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PID
-> FROM Batting
-> WHERE MatchID = 2689
-> AND PID IN
-> (SELECT PID
-> FROM Bowling
-> WHERE MatchID = 2689);
```

PID
99001
24001
23001
98002

4 rows in set (0.0022 sec)

Q5. Find Ids of all players that have either bowled or batted (or did both) in ODI match 2689

```
SELECT PID FROM Batting WHERE MatchID=2689 UNION SELECT PID FROM
Bowling WHERE MatchID=2689;
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PID FROM Batting WHERE MatchID=2689 UNION SELECT PID FROM Bowling WHERE MatchID=2689;
```

PID
89001
23001
25001
99002
95001
24001
99001
27001
98002

9 rows in set (0.0020 sec)

Q6. Find Ids

```
SELECT PID FROM Batting WHERE MatchID=2689
AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PID
-> FROM Batting
-> WHERE MatchID = 2689
-> AND PID NOT IN
-> (SELECT PID
-> FROM Bowling
-> WHERE MatchID = 2689);
```

PID
89001
25001
99002
95001
27001

5 rows in set (0.0017 sec)

Q7. Find match IDs where Tendulkar has played.

```
SELECT MatchID FROM Batting,
```

```
(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE  
PID=ST.PlayerID;
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT MatchID  
-> FROM Batting,  
-> (SELECT PlayerID  
-> FROM Player  
-> WHERE LName = 'Tendulkar') ST  
-> WHERE PID = ST.PlayerID;  
  
+-----+  
| MatchID |  
+-----+  
| 2689 |  
+-----+  
1 row in set (0.0084 sec)
```

Q8(a). Find Match IDs where player 27001 scored more runs than he made at every match played at Colombo.

```
SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns>ALL (SELECT NRuns  
FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM  
ODIMatch WHERE Ground='Colombo'));
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT MatchID  
-> FROM Batting  
-> WHERE PID=27001  
-> AND NRuns > ALL (  
-> SELECT NRuns  
-> FROM Batting  
-> WHERE PID=27001  
-> AND MatchID IN (  
-> SELECT MatchID  
-> FROM ODIMatch  
-> WHERE Ground='Colombo'  
-> )  
-> );  
  
+-----+  
| MatchID |  
+-----+  
| 2689 |  
+-----+  
1 row in set (0.0065 sec)
```

Q8(b). Find Match IDs where player 27001 scored fewer runs than he made at every match played at Brisbane.

```
SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns<ALL (SELECT NRuns  
FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM  
ODIMatch WHERE Ground='Brisbane'));
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT MatchID  
-> FROM Batting  
-> WHERE PID=27001  
-> AND NRuns < ALL (  
-> SELECT NRuns  
-> FROM Batting  
-> WHERE PID=27001  
-> AND MatchID IN (  
-> SELECT MatchID  
-> FROM ODIMatch  
-> WHERE Ground='Brisbane'  
-> )  
-> );  
  
+-----+  
| MatchID |  
+-----+  
| 2755 |  
+-----+  
1 row in set (0.0025 sec)
```

Q9(a). Find IDs and scores of players who scored less than 75 but more than 50 in Colombo.

```
SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND
MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PID, NRuns
-> FROM Batting
-> WHERE NRuns BETWEEN 51 AND 74
-> AND MatchID IN (
-> SELECT MatchID
-> FROM ODIMatch
-> WHERE Ground='Colombo'
-> );
```

PID	NRuns
25001	71
91001	60

2 rows in set (0.0018 sec)

Q9(b). Find IDs and scores of players who scored greater than 49 runs in Colombo.

```
SELECT PID,NRuns FROM Batting WHERE NRuns>49 AND MatchID IN (SELECT
MatchID FROM ODIMatch WHERE Ground='Colombo');
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PID, NRuns
-> FROM Batting
-> WHERE NRuns>49
-> AND MatchID IN (
-> SELECT MatchID
-> FROM ODIMatch
-> WHERE Ground='Colombo'
-> );
```

PID	NRuns
25001	71
91001	60

2 rows in set (0.0046 sec)

Q10. Find Player IDs whose date of first match (FTest) is not given.

```
SELECT PlayerID FROM Player WHERE FTest IS NULL;
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PlayerID
-> FROM Player
-> WHERE FTest IS NULL;
```

PlayerID
27001

1 row in set (0.0011 sec)

Q11. Find names of players who batted in match 2689.

```
SELECT FName,LName FROM Player WHERE EXISTS (SELECT * FROM Batting
WHERE PlayerID=PID AND MatchID=2689);
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT FName, LName
-> FROM Player
-> WHERE EXISTS (
->   SELECT *
->   FROM Batting
->   WHERE PlayerID=PID
->   AND MatchID=2689
-> );
```

FName	LName
Sachin	Tendulkar
Yuvraj	Singh
MS	Dhoni
Adam	Gilchrist
Ricky	Ponting
Andrew	Symonds
Breet	Lee
Praveen	Kumar
Harbhajan	Singh

9 rows in set (0.0029 sec)

Q12. Find Player IDs of players who have scored more than 30 in every ODI match they have played.

```
SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM
Batting b2 WHERE b1.PID=b2.PID AND b2.NRuns<31);
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PID AS PlayerID
-> FROM Batting b1
-> WHERE NOT EXISTS (
->   SELECT *
->   FROM Batting b2
->   WHERE b1.PID=b2.PID
->   AND b2.NRuns<31
-> );
```

PlayerID
24001
25001
25001
89001
91001

5 rows in set (0.0018 sec)

Q13. Find names of players that have batted in all ODI matches in Melbourne.

```
SELECT FName,LName FROM Player WHERE NOT EXISTS (SELECT * FROM
ODIMatch WHERE Ground='Melbourne' AND NOT EXISTS (SELECT * FROM Batting
WHERE PID=PlayerID AND Batting.MatchID=ODIMatch.MatchID));
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT FName, LName
-> FROM Player
-> WHERE NOT EXISTS (
->   SELECT *
->   FROM ODIMatch
->   WHERE Ground='Melbourne'
->   AND NOT EXISTS (
->     SELECT *
->     FROM Batting
->     WHERE PID=PlayerID
->     AND Batting.MatchID=ODIMatch.MatchID
->   )
-> );
```

Empty set (0.0051 sec)

```
MySQL localhost:33060+ ssl aarya61_match SQL >
```

Q14. Find the number of players who bowled in ODI match 2689.

```
SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;
+-----+
| NBowlers |
+-----+
|         4 |
+-----+
1 row in set (0.0035 sec)
```

Q15. Find the average batting score of all players who batted in ODI match 2689.

```
SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT AVG(NRuns) AS AvgRuns_2689
-> FROM Batting
-> WHERE MatchID=2689;
+-----+
| AvgRuns_2689 |
+-----+
|       25.2222 |
+-----+
1 row in set (0.0044 sec)
```

Q16. Find the youngest player in the database.

```
SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT
MAX(YBorn) FROM Player);
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);
+-----+
| Youngest_Player |
+-----+
| Ishant          |
+-----+
1 row in set (0.0060 sec)
```

Q16. Find the youngest player in the database.

```
SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT
MAX(YBorn) FROM Player);
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT Country,COUNT(*) AS NPlayers FROM Player GROUP BY Country;
+-----+-----+
| Country | NPlayers |
+-----+-----+
| India   |        10 |
| Australia |         5 |
| West Indies |         1 |
| Sri Lanka |         3 |
| Pakistan |         2 |
| Zimbabwe |         1 |
| South Africa |         4 |
| New Zealand |         2 |
| England |         1 |
+-----+-----+
9 rows in set (0.0016 sec)
```

Q17. Find the number of players from each country.

```
SELECT Country,COUNT(*) AS NPlayers FROM Player GROUP BY Country;
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT Country, COUNT(*) AS NPlayers
-> FROM Player
-> GROUP BY Country;
```

Country	NPlayers
India	10
Australia	5
West Indies	1
Sri Lanka	3
Pakistan	2
Zimbabwe	1
South Africa	4
New Zealand	2
England	1

9 rows in set (0.0020 sec)

Q18. Find batting average of each player. Present PID of each player.

SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PID, AVG(NRuns) AS AVGR
-> FROM Batting
-> GROUP BY PID;
```

PID	AVGR
23001	19.0000
24001	42.0000
25001	53.5000
27001	4.5000
89001	91.0000
91001	60.0000
92002	1.0000
94002	17.0000
95001	1.0000
98002	2.5000
99001	7.0000
99002	2.0000

12 rows in set (0.0017 sec)

Q19. Find batting average of each player from India. Present first name of each player.

SELECT FName,AVG(NRuns) AS Batting_Average FROM Player,Batting WHERE PlayerID=PID AND Country='India' GROUP BY PlayerID,FName;

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT FName, AVG(NRuns) AS Batting_Average
-> FROM Player, Batting
-> WHERE PlayerID=PID
-> AND Country='India'
-> GROUP BY PlayerID, FName;
```

FName	Batting_Average
Yuvraj	19.0000
MS	53.5000
Praveen	4.5000
Sachin	91.0000
Harbhajan	2.5000

5 rows in set (0.0049 sec)

Q20. Find the average score for each player when playing against Australia.

```
SELECT PID,AVG(NRuns) AS Avg_Score FROM Batting,ODIMatch WHERE
Batting.MatchID=ODIMatch.MatchID AND (Team1='Australia' OR Team2='Australia')
GROUP BY PID;
```

```
MySQL localhost:33060+ ssl aarya61_match SQL> SELECT PID, AVG(NRuns) AS Avg_Score
-> FROM Batting, ODIMatch
-> WHERE Batting.MatchID=ODIMatch.MatchID
-> AND (Team1='Australia' OR Team2='Australia')
-> GROUP BY PID;
```

PID	Avg_Score
89001	91.0000
23001	38.0000
25001	36.0000
99002	2.0000
95001	1.0000
24001	42.0000
99001	7.0000
27001	7.0000
98002	3.0000

9 rows in set (0.0016 sec)

Q21. Find the IDs of players who had a higher average score than all players when they played against Sri Lanka.

```
SELECT b1.PID FROM Batting b1,ODIMatch o1 WHERE b1.MatchID=o1.MatchID AND
(o1.Team1='Sri Lanka' OR o1.Team2='Sri Lanka') GROUP BY b1.PID HAVING
AVG(b1.NRuns)>ALL (SELECT AVG(b2.NRuns) FROM Batting b2,ODIMatch o2 WHERE
b2.MatchID=o2.MatchID AND (o2.Team1='Sri Lanka' OR o2.Team2='Sri Lanka') AND
b2.PID<>b1.PID GROUP BY b2.PID);
```

```
MySQL localhost:33060+ ssl aarya61_match SQL> SELECT b1.PID
-> FROM Batting b1, ODIMatch o1
-> WHERE b1.MatchID=o1.MatchID
-> AND (o1.Team1='Sri Lanka' OR o1.Team2='Sri Lanka')
-> GROUP BY b1.PID
-> HAVING AVG(b1.NRuns) > ALL
-> (
-> SELECT AVG(b2.NRuns)
-> FROM Batting b2, ODIMatch o2
-> WHERE b2.MatchID=o2.MatchID
-> AND (o2.Team1='Sri Lanka' OR o2.Team2='Sri Lanka')
-> AND b2.PID<>b1.PID
-> GROUP BY b2.PID
-> );
```

PID
25001

1 row in set (0.0067 sec)